

In class activity

STAT 720 | Summer 2026

2026-06-30

You are working with a collaborator in planning a new greenhouse experiment to answer the research question “do plant growth promoting bacteria increase crop yields?”. You are currently interested in testing 8 different bacteria, plus a control. Your response is yield (measured in $kg\ m^{-2}$).

You have data from a previous experiment, similar to the one you’re envisioning to answer this research question, that had a 2×3 factorial treatment structure and was run in an RCBD with 5 repetitions. Results from this experiment were $\hat{\sigma}_{\varepsilon}^2 = 5\ (kg\ m^{-2})^2$ and $\hat{\sigma}_{block}^2 = 1\ (kg\ m^{-2})^2$. No “significant differences” were found in terms of yield (measured in $kg\ m^{-2}$).

How many repetitions do you need now to obtain a precision that gives you a confidence interval that is at least $\hat{\theta} \pm 2\ kg\ m^{-2}$?